

GIANLUCA MONACI

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PROFESSIONAL EXPERIENCE

Naver Labs Europe

Senior Scientist, Project Lead

Sep 2017 - Present

Grenoble (FR)

- Research in Embodied AI, with focus end-to-end (RL) learning and vision-based approaches for precise, agile, robust real-world robot navigation
- Lead R&D projects and teams (up to 15 PhDs and engineers) developing algorithms and systems for autonomous robot navigation and data analytics

Xerox Research Center Europe

Laboratory Manager

Feb 2016 - Sep 2017

Grenoble (FR)

- Definition of data analytics research agendas, R&D portfolio and partnerships
- Management of research teams (up to 30 PhDs and engineers), program management

Philips Research

Senior Scientist, Project Lead

Nov 2008 - Feb 2016

Eindhoven (NL)

- Management of data analytics projects, including *HueLabs* concept testing platform and iF award-winning data analysis and visualization tools *Visual Retailer*
- Development of computer vision and data analytics solutions for lighting and healthcare
- Lead Philips contribution in several trans-organizational projects, including EU-funded projects and *Data Analytics for Lighting* Philips-MIT strategic partnership
- Creation, supervision and mentoring of data science teams

EDUCATION

University of California Berkeley, United States

PostDoc at Redwood Center for Theoretical Neuroscience

2007 - 2008

Swiss Federal Institute of Technology Lausanne (EPFL), Switzerland

Ph.D. in Signal, Image and Multi-modal data processing

2003 - 2007

University of Siena, Italy

Master's in Telecommunications Engineering

1996 - 2002

AWARDS

- **Best Paper Award finalist**, Robotics: Science and Systems (RSS) 2022
- **Outstanding Achievement Award** Philips Research 2015
- **iF Professional Concept Award** and **iF Communication Design Award** 2015 for the project Visual Retailer
- **Swiss National Science Foundation Fellowship for Advanced Researcher** 2008 (declined)
- **Swiss National Science Foundation Fellowship for Prospective Researcher** 2007
- **IBM Best Student Paper Award**, International Conference Image Processing (ICIP) 2005

OTHER PROFESSIONAL ACTIVITIES

- **Expert Evaluator** for the European Research Executive Agency, 2019 – Present
- **Exhibits and Demo Chair** at the International Conference on Image Processing (ICIP) 2022
- **Organizer** of the 2nd Naver Labs Europe International Workshop on AI for Robotics 2021
- **Development of large interactive light installations** at STRP Festival Eindhoven 2010-2011, and GLOW 2013

SKILLS

Research	Robot Navigation, Embodied AI, Computer Vision, Machine Learning, Multimodal data processing
Technical	Python, PyTorch, ROS, 3D simulators (AiHabitat, Gibson, Isaac)
Leadership	Technology transfer, Management of R&D programs, teams, projects, partnerships
Languages	Italian, English, French, Spanish, Dutch

PUBLICATIONS

2026

Compressing Observation History into Agent Memory: Distilling Transformers into Recurrent Transformers
Philippe Weinzaepfel, Christian Wolf, Mert Bülent Sariyıldız, Guillaume Bono, **Gianluca Monaci**
arXiv preprint arXiv:2606.21562

Advantage-Driven Explicit Memory for Social Navigation
Yeonsoo Park, Mattia Racca, Guillaume Bono, Steeven Janny, **Gianluca Monaci**, Tomi Silander, Christian Wolf
International Conference on Robotics & Automation (ICRA) 2026 SCR Workshop

What does really matter in image goal navigation? (Oral)
Gianluca Monaci, Philippe Weinzaepfel, Christian Wolf
International Conference on 3D Vision (3DV) 2026

2025

RANa: Retrieval-Augmented Navigation
Gianluca Monaci, Rafael S. Rezende, Romain Deffayet, Gabriela Csurka, Guillaume Bono, Hervé Déjean, Stéphane Clinchant, Christian Wolf
Transactions on Machine Learning Research (TMLR) 2025

Kinaema: a recurrent sequence model for memory and pose in motion
Mert Bülent Sariyıldız, Philippe Weinzaepfel, Guillaume Bono, **Gianluca Monaci**, Christian Wolf
The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS) 2025

Robots waiting for the elevator: Integrating social norms in a low-data regime goal selection problem
Mattia Racca, Jutta Willamowski, Tommaso Colombino, **Gianluca Monaci**, Danilo Gallo
International Conference on Robot and Human Interactive Communication (RO-MAN) 2025

Reasoning in visual navigation of end-to-end trained agents: A dynamical systems approach (Highlight)
Steeven Janny, Hervé Poirier, Leonid Antsfeld, Guillaume Bono, **Gianluca Monaci**, Boris Chidlovskii, Francesco Giuliani, Alessio Del Bue, Christian Wolf
Computer Vision and Pattern Recognition (CVPR) 2025

2024

Zero-BEV: Zero-shot projection of any first-person modality to BEV maps (Spotlight)
Gianluca Monaci, Leonid Antsfeld, Boris Chidlovskii, Christian Wolf
International Conference on 3D Vision (3DV) 2024

Learning to navigate efficiently and precisely in real environments
Guillaume Bono, Hervé Poirier, Leonid Antsfeld, **Gianluca Monaci**, Boris Chidlovskii, Christian Wolf
Computer Vision and Pattern Recognition (CVPR) 2024

Learning with a Mole: Transferable latent spatial representations for navigation without reconstruction
Guillaume Bono, Leonid Antsfeld, Assem Sadek, **Gianluca Monaci**, Christian Wolf
International Conference on Learning Representations (ICLR) 2024

2023

Learning whom to trust in navigation: Dynamically switching between classical and neural planning
Sombit Dey, Assem Sadek, **Gianluca Monaci**, Boris Chidlovskii, Christian Wolf
International Conference on Intelligent Robots and Systems (IROS) 2023

2022

DiPCAN: Distilling privileged information for crowd-aware navigation (Best Paper Finalist)
Gianluca Monaci, Michel Aractingi, Tomi Silander
Robotics: Science and Systems (RSS) 2022

2014 and earlier

Semantic video scene segmentation and transfer
Tommaso Gritti, Chris Damkat, **Gianluca Monaci**
Computer Vision and Image Understanding, 122, pages 172-181, 2014

LumaFluid: A responsive environment to stimulate social interaction in public spaces

Gianluca Monaci, Tommaso Gritti, Martine van Beers, Ad Vermeulen, Bram Nab, Inge Thomassen, Marigo Heijboer, Sandra Suijkerbuijk, Wouter Walmink, Maarten Hendriks
International Joint Conference on Ambient Intelligence (AmI) 2012

Indoor user zoning and tracking in passive infrared sensing systems

Gianluca Monaci, Ashish Pandharipande
European Signal Processing Conference (EUSIPCO) 2012

Automatic light scene setting through image-based sparse light effect approximation

Tommaso Gritti, **Gianluca Monaci**
IEEE Transactions on Multimedia, 14(4), pages 1351-1358, 2012

ImagiLight: A vision approach to lighting scene setting

Tommaso Gritti, **Gianluca Monaci**
ACM Multimedia 2011

Flower power

Gianluca Monaci, Tommaso Gritti, Fabio Vignoli, Wouter Walmink, Maarten Hendriks
ACM Multimedia 2011

Ultrasound video analysis for understanding infant breastfeeding

Gianluca Monaci, Mike Woolridge
International Conference on Image Processing (ICIP) 2011

Towards real-time audiovisual speaker localization

Gianluca Monaci
European Signal Processing Conference (EUSIPCO) 2011

Blind audiovisual source separation based on sparse redundant representations

Anna Llagostera Casanovas, **Gianluca Monaci**, Pierre Vanderghenst, Rémi Gribonval
IEEE Transactions on Multimedia, 12(5), pages 358-371, 2010

Learning bimodal structure in audio-visual data

Gianluca Monaci, Pierre Vanderghenst, Friedrich T. Sommer
IEEE Transactions on Neural Networks, 20(12), pages 1898-1910, 2009

Geometric video approximation using weighted Matching Pursuit

Òscar Divorra Escoda, **Gianluca Monaci**, Rosa M. Figueras i Ventura, Pierre Vanderghenst, Michel Bierlaire
IEEE Transactions on Image Processing, 18(8), pages 1703-1716, 2009

Learning sparse generative models of audiovisual signals

Gianluca Monaci, Friedrich T. Sommer, Pierre Vanderghenst
European Signal Processing Conference (EUSIPCO) 2008

Blind audiovisual separation based on redundant representations

Anna Llagostera Casanovas, **Gianluca Monaci**, Pierre Vanderghenst, Rémi Gribonval
International Conference on Acoustics, Speech, and Signal Processing (ICASSP) 2008

Blind audiovisual source separation using sparse representations

Anna Llagostera Casanovas, **Gianluca Monaci**, Pierre Vanderghenst
International Conference on Image Processing (ICIP) 2008

Learning multimodal dictionaries

Gianluca Monaci, Philippe Jost, Pierre Vanderghenst, Boris Mailhe, Sylvain Lesage, Rémi Gribonval
IEEE Transactions on Image Processing, 16(9), pages 2272-2283, 2007

Tracking Atoms with particles for audio-visual source localization

Gianluca Monaci, Pierre Vanderghenst, Emilio Maggio, Andrea Cavallaro
International Conference on Acoustics, Speech, and Signal Processing (ICASSP) 2007

Analysis of multimodal sequences using geometric video representations

Gianluca Monaci, Òscar Divorra Escoda, Pierre Vanderghenst
Signal Processing, 86(12), pages 3534-3548, 2006

Learning multi-modal dictionaries: Application to audiovisual data

Gianluca Monaci, Philippe Jost, Pierre Vanderghelynst, Boris Mailhe, Sylvain Lesage, Rémi Gribonval
International Workshop on Multimedia Content Representation, Classification and Security, 2006

Audiovisual Gestalts

Gianluca Monaci, Pierre Vanderghelynst

Computer Vision and Pattern Recognition Workshop (CVPRW) 2006

Analysis of multimodal signals using redundant representations (Best Student Paper)

Gianluca Monaci, Òscar Divorra Escoda, Pierre Vanderghelynst

International Conference on Image Processing (ICIP) 2005

Learning structured dictionaries for image representation

Gianluca Monaci, Pierre Vanderghelynst

International Conference on Image Processing (ICIP) 2004

Image compression with learnt tree-structured dictionaries

Gianluca Monaci, Philippe Jost, Pierre Vanderghelynst

International Workshop on Multimedia Signal Processing (MMSP) 2004

Chromatic contrast detection in spatial chromatic noise

Gianluca Monaci, Gloria Menegaz, Sabine Süssstrunk, Kenneth Knoblauch

Visual Neuroscience, 21(3), pages 291-294, 2004

PATENTS AND PATENT APPLICATIONS

System and method for generating a bird-eye view map

Gianluca Monaci, Leonid Antsfeld, Boris Chidlovskii, Christian Wolf

US20250265778A1

Systems and methods for spatial representations for navigation without reconstruction

Guillaume Bono, Leonid Antsfeld, **Gianluca Monaci**, Assem Sadek, Christian Wolf

US20240408756A1

Distilling information for crowd-aware robot navigation

Gianluca Monaci, Michel Aractingi, Tomi Silander

US20230359212A1

Method and system for generating a schematic map

Gianluca Monaci, Frédéric Roulland, Adrien Bruyat, Christelle Loiodice

US20230394399A1

Lighting commissioning system using position sensitive detectors

Harry Broers, Tommaso Gritti, **Gianluca Monaci**

WO2015104603A2

Sensor network with adaptive detection settings based on the status information from neighboring luminaires and/or connected devices

Gianluca Monaci, Remco Muijs, Harry Broers, Petrus Kuijten, Daniel Görgen, Tim Schenk, Muhammad Shoaib

WO2015028921A1

A video analysis device and a method of operating a video analysis device

Tommaso Gritti, **Gianluca Monaci**, Franck Van Heesch

WO2014180695A1

Light device controller, lighting device, and method of controlling light settings of a lighting device

Harry Broers, **Gianluca Monaci**, Tommaso Gritti

WO2014118675A1

A method of self-calibrating a lighting device and a lighting device performing the method

Gianluca Monaci, Oleg Belik, Tommaso Gritti, Petrus Kuijten

WO2014053954A1

Processing module for use in a presence sensing system

Harry Broers, Tommaso Gritti, **Gianluca Monaci**

WO2013179175A1

Vision system comprising an image sensor and means for analysis and reducing loss of illumination towards periphery of the field of view using multiple frames

Tommaso Gritti, **Gianluca Monaci**, Harry Broers

WO2013124772A1

Passive infrared sensor system for position detection

Gianluca Monaci, Ashish Pandharipande

WO2013128307A1

Monitoring a scene

Gianluca Monaci, Tommaso Gritti, Harry Broers

WO2013093771A1

Lighting control system

Gianluca Monaci, Tommaso Gritti

WO2012085742A1

Adaptive illumination

Gianluca Monaci, Tommaso Gritti

WO2012137046A1

Method and apparatus for generating a predetermined type of ambient lighting

Gianluca Monaci, Tommaso Gritti

WO2012001588A1